

Assessment of residential structures using UAV - A Case Study at Uppada, Andhra Pradesh

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Abstract

Communities along the low-lying coasts are increasingly vulnerable to extreme climatic events such as cyclones, storm surges, and sea level rise, which pose significant risks to life, infrastructure, and livelihoods. The advancement of high-resolution geospatial data has enabled the monitoring of coastal areas and the provision of valuable synoptic information for disaster risk assessment. This study assesses storm surge-induced flood exposure and structural susceptibility of coastal residential structures is based on a static inundation approach at Uppada village, Andhra Pradesh, India, using high-resolution UAV-derived geospatial datasets. A total of 1,327 buildings were extracted from orthomosaic images and classified according to their construction typology. These buildings were then analysed for their potential flood exposure impact based on a storm surge scenario, considering return periods of 10 years, 50 years, and 100 years, as adapted from a previous modelling study. The results reveal that flood exposure increases sharply with rising storm intensity; only 7 buildings (~0.5%) are located close to near-threshold categories under the 10-year return period, while nearly 397 buildings (30%) are inundated under the 100-year return period. A deeper analysis of inundation depth in relation to different building types reveals important insights for disaster preparedness. Structures such as huts and asbestos sheds are significantly more vulnerable to flooding compared to concrete buildings, which demonstrate a high level of resilience. These results demonstrate the potential of UAV-based geospatial approaches to support data-driven decision-making for sustainable coastal management and climate resilience. Furthermore, UAV-based vulnerability mapping enables the identification of the most vulnerable infrastructure and habitation in low-lying areas. This will assist the local administration and disaster management authorities as a critical decision-support tool for formulating immediate preparedness plans to evacuate and relocate residents in these areas, which are highly vulnerable during extreme flooding events.

Keywords: UAV, Storm Surge, Flood Inundation Depth, Building Vulnerability Assessment, Climate Change

59 **1 INTRODUCTION**

60 With the accelerating pace of urbanisation in low-lying coastal regions, more than one billion people living in
61 coastal cities and settlements are projected to be exposed to coast-specific climate hazards by 2050
62 (Intergovernmental Panel On Climate Change (Ipcc) 2022, 2023) which pose significant risks to life,
63 infrastructure, and livelihoods. The vulnerability to the hazards in the coastal zone is being assessed by system
64 susceptibility, resilience, and resistance, allowing adaptive management plans to be developed in perspective with
65 the latest Intergovernmental Panel on Climate Change(IPCC) report (Noor and Abdul Maulud 2022).

66 Along the Indian coastline, climate change-induced tropical cyclones pose a persistent threat, often resulting in
67 widespread destruction through storm surges (Rao et al. 2020) and coastal inundation (Diaz et al. 2024; Bharadwaz
68 et al. 2025). The North Indian Ocean alone generates nearly 13% of the world's tropical cyclones, leaving India's
69 coastal regions—particularly the east coast, which experiences more frequent cyclonic activity than the west—
70 highly exposed to future extreme events (Rao et al. 2020). It could negatively impact the coastal regions of the
71 country with severe coastal flooding, shoreline erosion (Lloyd et al. 2016; Chao et al. 2020; Rao et al. 2020; Fang
72 et al. 2023), sedimentation, and saline water intrusion (both ground and surface) (Rezaie et al. 2019). In the last
73 decade, major tropical cyclones, including Phailin (2013), Hudhud (2014), Vardah (2016), Titli and Phethai
74 (2018), Fani (2019), Amphan (2020), Yaas (2021), and Asani (2022), have caused extensive damage along India's
75 east coast (Bharadwaz et al. 2025), leading to considerable economic losses, displacement, and loss of life.
76 Cyclone Phailin alone resulted in economic losses of approximately USD 174 million (Padhy et al. 2015),
77 underscoring that India's coastal regions remain inadequately prepared to cope with cyclonic hazards (Bharadwaz
78 et al. 2025)

79 The National Disaster Management Authority (NDMA) of India identifies storm surges associated with tropical
80 low-pressure systems as one of the most destructive hazards for coastal areas. When storm surges coincide with
81 high astronomical tides, the resulting storm tide significantly amplifies coastal flooding, producing even greater
82 risks to coastal settlements (Rao et al. 2020). These surges, primarily driven by intense winds pushing seawater
83 toward the coast, elevate sea levels far beyond normal tidal ranges and can extend inland through river networks,
84 inundating low-lying zones (Ganaie et al. 2025). At the same time, storm-induced waves, coupled with locally
85 generated winds, alter nearshore dynamics, triggering sediment loss, coastal erosion, and sea-level anomalies in
86 the form of storm surges (Noujas et al. 2023). The vulnerability of these regions is further exacerbated by densely
87 populated settlements, rapid urban expansion, poverty, and inadequate infrastructure planning (Chao et al. 2020).

Accurately assessing the coastal flood exposure to coastal structures requires a high-resolution representation of coastal topography and built infrastructure. While flood exposure is often defined through floodplain delineation that intersects with built assets, a lack of information on building-specific characteristics limits the accurate assessment of whether these assets are genuinely vulnerable to flooding (Diaz et al. 2024). Conventional field-based surveying approaches are inherently resource-intensive and time-consuming, and are further constrained by limited site access and the reduced availability or operational usability of high-resolution satellite imagery due to persistent cloud cover and severe atmospheric conditions. However, the emergence of advanced scanning and remote sensing technologies has revolutionised data collection, enabling the rapid acquisition of high-resolution geospatial datasets that can be efficiently used to derive building elevation and structural vulnerability information (Diaz et al. 2024). Moreover, commonly used coarse-resolution digital elevation models often fail to capture fine-scale topographic variability and building-level details essential for local-scale flood assessment.

Recent advances in unmanned aerial vehicle (UAV)-based photogrammetry have provided new opportunities for high-resolution coastal mapping. UAVs enable rapid acquisition of centimetre-scale topographic data, offering detailed digital surface models (DSM) and digital terrain models (DTM) suitable for building-level flood exposure analysis. Unlike satellite platforms, UAVs can be deployed flexibly during pre- and post-disaster phases and are less affected by cloud cover, making them particularly valuable for coastal hazard assessment and disaster risk reduction. Several recent studies (Chao et al. 2020; Diaz et al. 2024) have demonstrated the effectiveness of UAV-derived datasets for flood mapping, building damage assessment, and coastal vulnerability studies, highlighting their potential for local-scale decision support.

Despite these advancements, building-scale assessments of storm surge-induced flooding remain limited in many developing coastal regions, including India. Existing studies often focus on regional-scale hazard mapping or rely on simplified exposure metrics, with limited consideration of structural characteristics and fine-scale elevation differences that control inundation patterns. As a result, local administrations frequently lack detailed, spatially explicit information required to prioritise disaster preparedness, evacuation planning, and mitigation measures at the community level.

In this context, the present study aims to assess storm surge-induced flood exposure of coastal residential structures at Uppada village, Andhra Pradesh, using high-resolution UAV-derived geospatial datasets. The study integrates UAV-based topographic mapping with storm surge scenarios corresponding to different return periods to evaluate building-level inundation patterns. By analysing flood exposure in relation to structural typology and

elevation, this work seeks to identify structures that are particularly susceptible to storm surge flooding and to provide actionable insights for coastal disaster preparedness and climate-resilient planning. The outcomes of this study contribute to demonstrating the applicability of UAV-based approaches as a data-driven decision-support tool for sustainable coastal management in cyclone-prone regions.

2 DATA AND METHODOLOGY

2.1 Study Area

The study was conducted in Uppada village, located north of Kakinada district on the east coast of India ($17^{\circ}4' - 17^{\circ}5' \text{ N}$; $80^{\circ}19' - 80^{\circ}20' \text{ E}$) Figs 1, covers an area of 1.47 km^2 with a population of 12,964, as per the 2011 census of India (<https://census.indiagis.org/census/map/index.jsp>). It is a highly eroding coastline (Kankara et al. 2015; Gireesh et al. 2023) that is vulnerable to cyclones and storm surges, which have affected the adjoining roads and a few buildings in the village over the years. The region experiences monsoons, with the South West Monsoon (June–September) accounting for nearly half of the annual rainfall and the North East Monsoon (October–December) supplying the remainder. Additionally, human interventions, such as port development, dredging, and sand mining, have altered natural coastal processes, thereby accelerating geomorphological changes (Noujas et al. 2023) in the region.

Historic events, such as the 1977 Diviseema cyclone, the 1996 Godavari cyclone, and Cyclone Titli in 2018, have caused severe loss of life, infrastructure damage, and shoreline erosion on a large scale, with over 600 acres of land throughout the Kakinada suburbs disappearing over the previous decades (Bharadwaz et al. 2025). Coastal protection measures, such as seawalls and gabions, were implemented over the years as temporary measures to prevent residential infrastructure from coastal inundation and subsequent erosion. However, these protection strategies have been implemented piecemeal and stand disintegrated currently, Figs 2, and the rows of houses and the temple visible in the past are now submerged due to the combined effects. These conditions serve as an excellent case study to study the storm surge–induced flood exposure to the coastal residential structures at their present state and act as a base assessment to enhance disaster preparedness and promote sustainable coastal management in the study area.



Fig. 1 Study Area at Uppada Coast, Kakinada District



Fig. 2 Field photographs illustrating the status of the coast along the study area.

2.2 Data Collection

2.2.1 Aerial Survey

The aerial survey was conducted in August 2024 using the SNAP-M rotorcraft Unmanned Aerial Vehicle (UAV), Figs 3, at Uppada village, Kakinada, Andhra Pradesh. It is a DGCA-certified nadir-looking drone for detailed

terrain and area coverage. It is fitted with an RTK-PPK module for precise geolocation and height determinations. Furthermore, it features a 24-megapixel Sony digital camera, which is electronically triggered by the UAV's autopilot system, designed to capture images at specific intervals and locations. The flight planning was meticulously carried out using the CD Fly software by CD Space Robotic Private Limited. UAV is flying at a speed of 10 m/s with a wind resistance of >8 m/s during the flight. To ensure the creation of a 3-dimensional topography, flights were maintained in a uniform grid pattern with a side and front overlap of 70%. This helped in obtaining high-resolution data to determine and classify the buildings in the study area. The details of the camera specification and parameters are explained in Table 1.

Parameters	Specification
Manufacturer & Model	CD Space Robotics Private Ltd., & SNAP-M
Weight & Category	2.7 kg – Small Category
Endurance	50+ mins
Max. Altitude	120 m (DGCA-approved Limit)
Wind Resistance	>8 m/s
Flight Mode	Dual, Automatic based on waypoints or wireless control.
Horizontal Accuracy	< 10 cm
Vertical Accuracy	< 20 cm
Payloads/Camera	24 MP Sony Camera, 6000 x 4000 pix resolution images.
Ground Sampling Distance (GSD)	3 cm
Overlap (Front & Side)	70%

Table 1 Specification of SNAP-M UAV

A total of 1,205 aerial images were obtained with a ground sampling distance (GSD) of 3 cm/pixel at a flight altitude of 120 m AGL from three flights over the study area. The RTK-enabled UAV system created highly precise orthomosaic, which were cross-referenced with classified outputs and field-based control points(Ruksajai et al. 2025). Pix4D Mapper was utilised, designed specifically for aerial photogrammetry applications (Shukla and Jain 2020; Türk et al. 2022; Veeravalli et al. 2023; Lovrinčević et al. 2025; Gond et al. 2025), to generate dense point cloud, Orthomosaics, Digital Elevation Model (DEM), and Digital Terrain Model of the region with a resolution of 0.028 cm/pixel, Figs 4.



Fig. 3 SNAP-M rotorcraft unmanned aerial vehicle (UAV) (left) and UAV RTK Base Station (right).

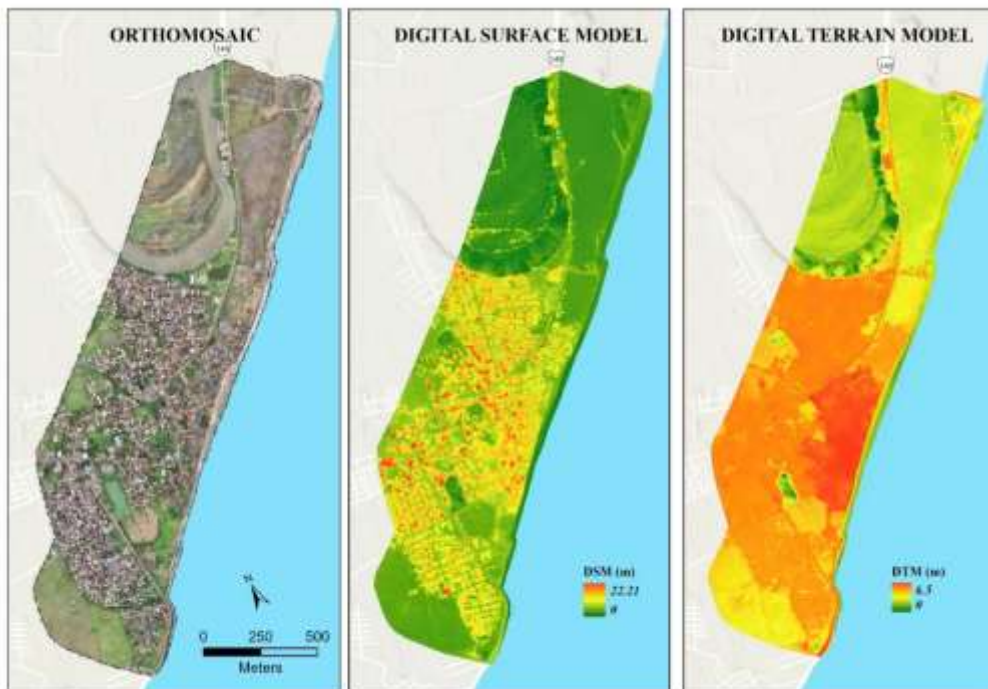


Fig. 4 UAV-generated orthomosaic, DSM, and DTM highlighting geomorphic and elevation variations along the Uppada coast.

2.2.2 Ground Control Points (GCPs)

Ground control points (GCPs), essential for orienting UAV imagery and ensuring data fidelity, were established through ground truthing using a real-time kinematic (RTK) approach (Ruksajai et al. 2025; Gond et al. 2025). For this study, positional data were acquired using a Leica GS18 T GNSS RTK rover, which combines an Inertial Measurement Unit (IMU) & GNSS to accurately collect data even when the rover pole is tilted up to ~300 degrees. A base station was set up within the study site using the static method, serving as a permanent GCP and ensuring

positional stability throughout the survey period (Gond et al. 2025). A total of 232 GCPs were established and categorised into three classes: beach and land areas, to accurately measure the terrain.

2.2.3 Evaluation of UAV and GCPs

Validation of UAVs equipped with in-built RTK/PPK modules against independently surveyed RTK ground control points (GCPs) is essential because onboard RTK/PPK solutions, while providing high-precision positioning, can still be affected by systematic biases, datum inconsistencies, or GNSS signal errors. Accuracy was evaluated using standard error metrics, including Mean Error (ME) for average bias, Standard Deviation of Errors (SDE) for variability, Root Mean Square Error (RMSE) for overall accuracy, and R^2 values (Clapuyt et al. 2016; Zhang et al. 2022). The statistical comparison between UAV-derived elevations and RTK-GNSS ground measurements demonstrated strong vertical accuracy across both surface types, with minor variations between the beach and land areas, Table 2. On the beach, the mean error (ME_z) was 0.187, indicating a slightly higher vertical deviation compared to the land, which recorded a mean error of 0.057. Similarly, the standard deviation (SD_z) was marginally higher for the beach (0.093) than for the land (0.085 m), reflecting greater variability in elevation measurements over sandy and irregular surfaces. The root mean square error (RMSE_z) followed a similar pattern, with values of 0.207 m for the beach and 0.101 m for the land, confirming higher precision over stable, well-defined land surfaces. Despite these differences, both environments exhibited excellent correlation between UAV and RTK data, with R^2 values of 0.989 for the beach and 0.993 for the land, signifying that the UAV-derived models provide highly reliable vertical accuracy suitable for detailed topographic analysis. The analysis revealed variations within a limiting range in the Z value depending on the surface characteristics of the region.

STATISTICAL VALIDATION (Z)	BEACH	LAND
Mean Error (ME _z)	0.187	0.057
Standard Deviation (SD _z)	0.093	0.085
RMSE _z	0.207	0.101
R^2_z	0.989	0.993

Table 2 Statistical Validation of RTK with UAV

2.4 Building Footprint Extraction and Structural Classification

Building footprints were manually digitised from orthomosaic images in ArcGIS Pro software to ensure accurate delineation of individual structures, resulting in a total of 1,327 buildings. Manual digitisation was adopted due to the complexity of roof structures and the presence of closely spaced buildings, which limited the reliability of

automated extraction techniques. Residential buildings were classified into structural typology based on construction type and rooftop material, comprising Concrete buildings (29.1%), Tile-roofed buildings (28.6%), Asbestos sheds (30.7%), and Huts (11.6%). This classification was based on visual interpretation of high-resolution imagery and field knowledge of local construction practices. The height of each building was estimated using the normalised Digital Surface Model (nDSM), derived from the difference between the DSM and DTM, a widely applied method for building height extraction (Shukla and Jain 2020). The digitised building footprints were converted to points, and the nDSM was masked to these footprints to extract height values above the ground level. Focal Statistics was then used to transfer height information to the attribute table, which was subsequently joined to the footprint points. The spatial distribution and classification of building footprints are presented in Figs 5.

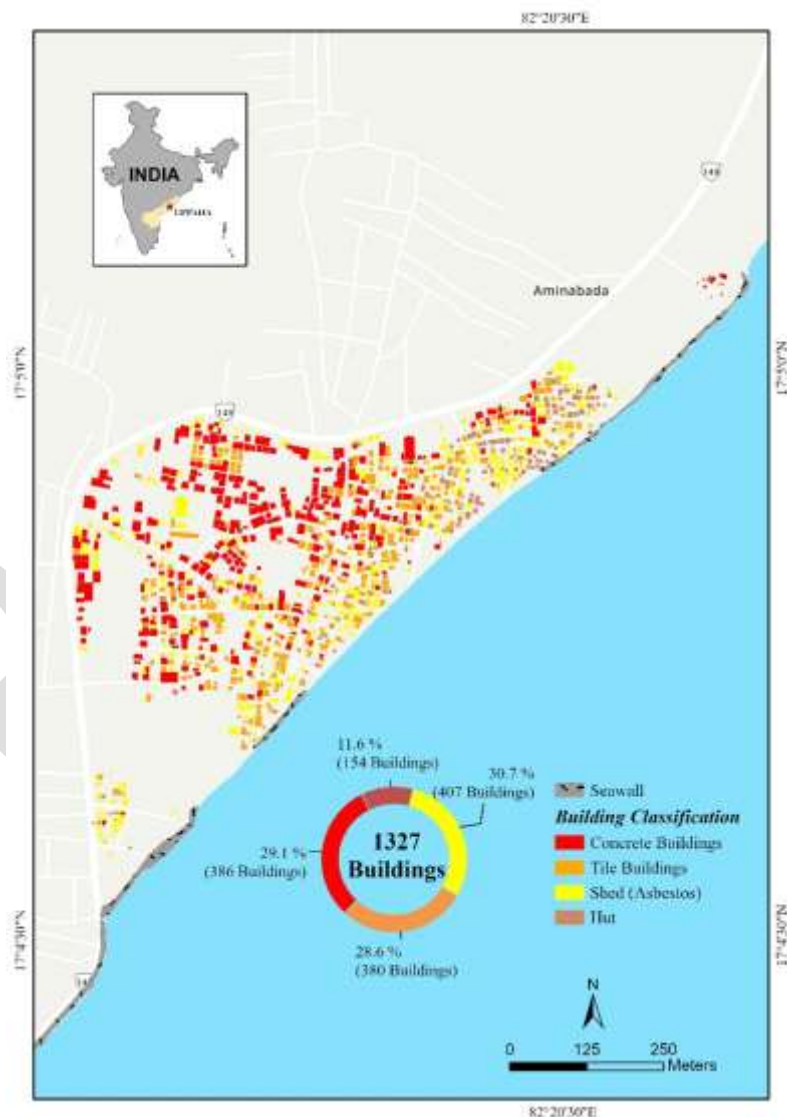


Fig. 5 Mapping of Building Classification

2.5 Storm Surge Scenarios and Flood Exposure Estimation

The maritime state on the southeastern coast of India is characterised by its vulnerability to tropical cyclones that strike the region with alarming frequency each year. These powerful storms primarily endanger coastal areas, which are low-lying and susceptible to inundation during extreme weather events, particularly the deltaic regions of the Krishna and Godavari rivers. In this study, the potential inundation of residential structures in Uppada was assessed by combining estimates of maximum water elevation (MWE) with model simulations of storm surge-induced flooding. As our study site is located on the northern side of the Godavari River, MWEs vary from 3.5m to 5.1 m (Rao et al. 2020). Floodwater depth is the height of the water level above ground at a specified location and is a function of the elevation of the region and the storm surge height. It indicates the maximum height to which inundated water can rise near or within the building during extreme events (Chao et al. 2020).

By adapting the methodology proposed by (Knutson et al. 2010) and (Rao et al. 2020) MWE values were applied to the Uppada coast under three distinct climatic scenarios, which involves slightly increasing cyclone wind speeds based on global and regional climatic projections, to understand coastal flooding due to climate change.

Flood inundation depth (FD) at each building location was estimated using the relation:

$$\text{Flood Inundation Depth (FD)} = \text{MWE} - \text{GE} \quad (1)$$

where MWE is the maximum water elevation for a given storm surge scenario, and GE is the ground elevation derived from the UAV-based DTM. Positive FD values indicate inundation, whereas negative values indicate buildings located above the projected surge level. This approach represents a static inundation (bathtub) method, assuming a uniform water level across the study area. While this method does not account for dynamic hydrodynamic processes such as wave setup, inland flow resistance, or drainage effects, it provides a first-order estimate of potential inundation extent suitable for screening-level exposure assessment.

2.6 Flood Inundation Depth Classification

Furthermore, the residential structures' exposure was categorised into flood exposure classes based on the estimated flood depth, following the threshold values adapted from (Diaz et al. 2024). The classification of structures is organised into six categories based on the positive and negative values of Flood Depth (FD) as presented in Table 3. These classes represent relative levels of flood exposure rather than direct damage estimates. The classification framework enables comparison of exposure across structural types and storm surge scenarios and supports identification of buildings requiring priority consideration for disaster preparedness and evacuation planning.

Flood Inundation Depth Classification for Buildings	
Major Inundation	$> 0.61m$
Moderate Inundation	$0.3m - 0.61m$
Minor Inundation	$0 - 0.29m$
At Risk	$-0.01m - -0.29m$
Safe	$-0.3m - -0.61mm$
Very Safe	$< -0.61m$

Table 3 Details of Flood depth classification for residential structures (Diaz et al. 2024)

Large Language Models (LLMs), including ChatGPT, were used in a limited capacity to support language refinement and clarity of the manuscript. The use of AI tools was restricted to AI-assisted copy editing, including improvements in readability, grammar, spelling, punctuation, and academic tone of human-generated text. No AI tool was used for data analysis, interpretation of results, generation of scientific content, or decision-making. The complete Methodology workflow is illustrated in Figs. 6.

3 RESULTS

3.1 Flood Exposure Analysis and Building Classification

The storm surge-induced flood exposure of residential buildings in Uppada was assessed under three scenarios corresponding to 10-year, 50-year, and 100-year return periods. Flood exposure was quantified using building-level flood depth estimates derived from the difference between maximum water elevation and ground elevation. Positive flood depth values indicate inundation, whereas negative values indicate buildings located above the projected surge level.

A total of 1,327 buildings were evaluated in the study area, Figs. 7 for different scenarios. Under the present condition (10-year return period) storm surge scenario of flooding level of 3.5m indicates, no buildings experience positive flood depths, indicating the absence of inundation across the study area. However, seven buildings (~0.5% of the total buildings) are located close to the projected surge elevation and fall within near-threshold categories. When considering the scenario, the 50-year return period of storm surge projected flood level of 4.6m, which involves a 7% increase in wind speed, it is estimated that 296 buildings are exposed to the flood level impact. Of these, 230 buildings (~17% of the total buildings) experience inundation, and the remaining 66 buildings are close to the projected surge level without affecting a significant portion of the built environment.

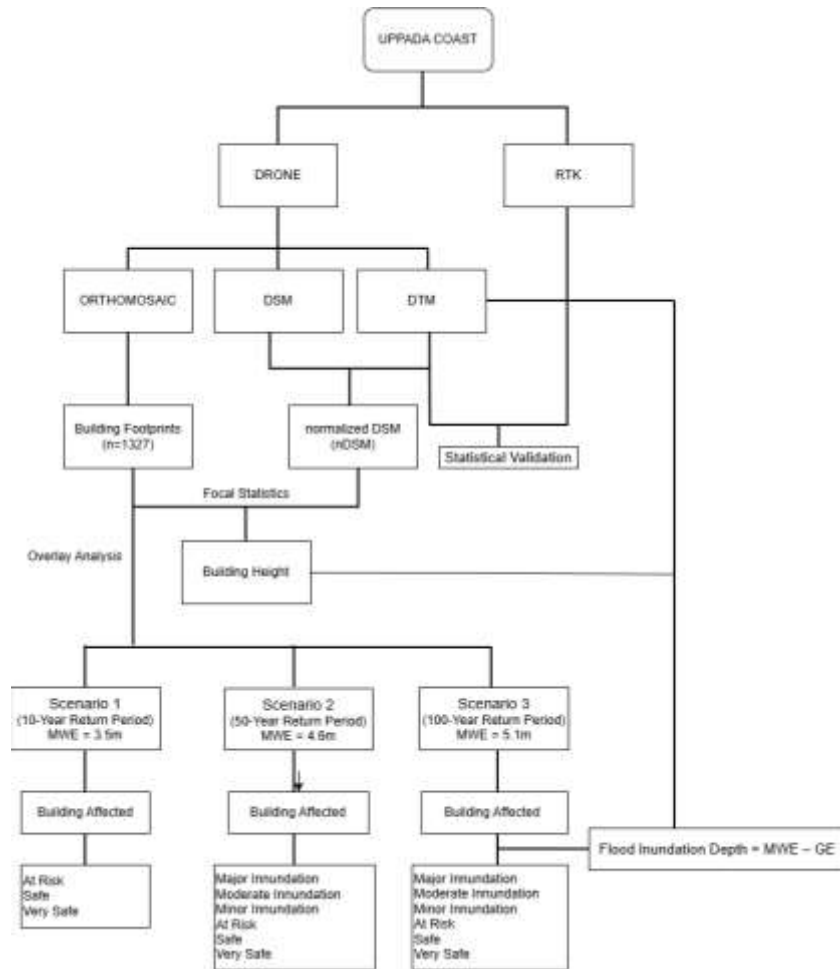


Fig. 6 Data collection and methodology workflow of the study.

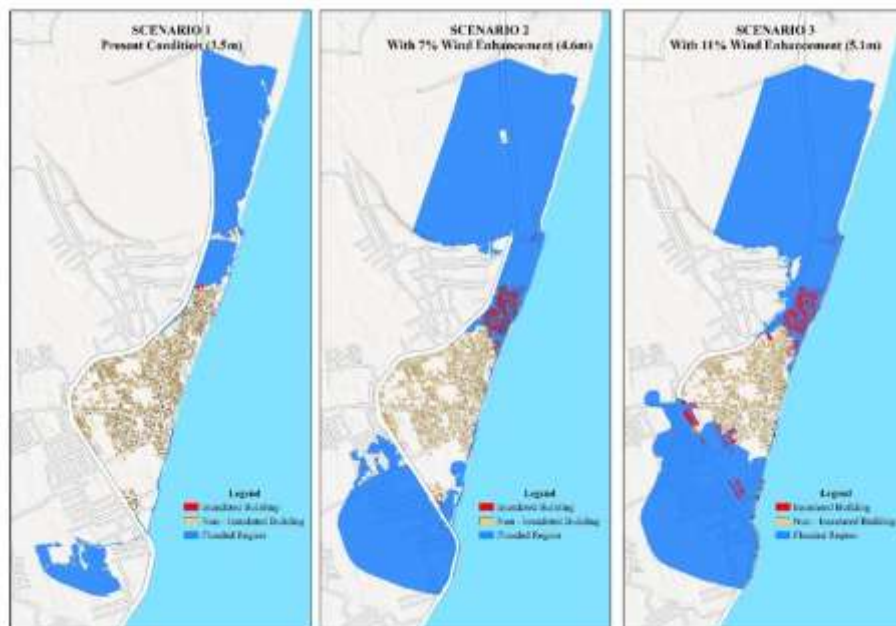


Fig. 7 Flood inundation impact on the study area for different climate scenarios



Fig. 8 Building Classification with Flood Inundation Depth

The extreme storm surge scenario corresponding to the 100-year return period results in widespread inundation across the study area, which has an 11% increase in wind intensity and a 5.1m increase in water level. A total of 408 residential buildings were exposed to the flood level threshold value. Among these, 192 structures fall into the major inundation category, 163 into the moderate category, and 42 into the minor inundation category, which is 397 buildings (30% of total buildings). Only 11 buildings remain above the projected surge level, comprising eight at risk and three safe structures, as illustrated in Table 4. The three-dimensional representation of Scenario 3 shows an 11% enhancement in wind conditions alongside the depth of flood inundation, as depicted in Figs. 9 for the entire study site.

Flood Inundation Depth Classification for Buildings		Present Condition (3.5m)	With 7% Wind Enhancement (4.6m)	With 11% Wind Enhancement (5.1 m)
Major Inundation	$> 0.61m$	0	26	192
Moderate Inundation	$0.3m - 0.61m$	0	58	163
Minor Inundation	$0 - 0.29m$	0	146	42
At Risk	$-0.01m - -0.29m$	3	59	8
Safe	$-0.3m - -0.61mm$	3	3	3
Very Safe	$< -0.61m$	1	4	0
Total		7	296	408

Table 4 Flood depth classification to storm surge scenarios in Uppada, Kakinada.

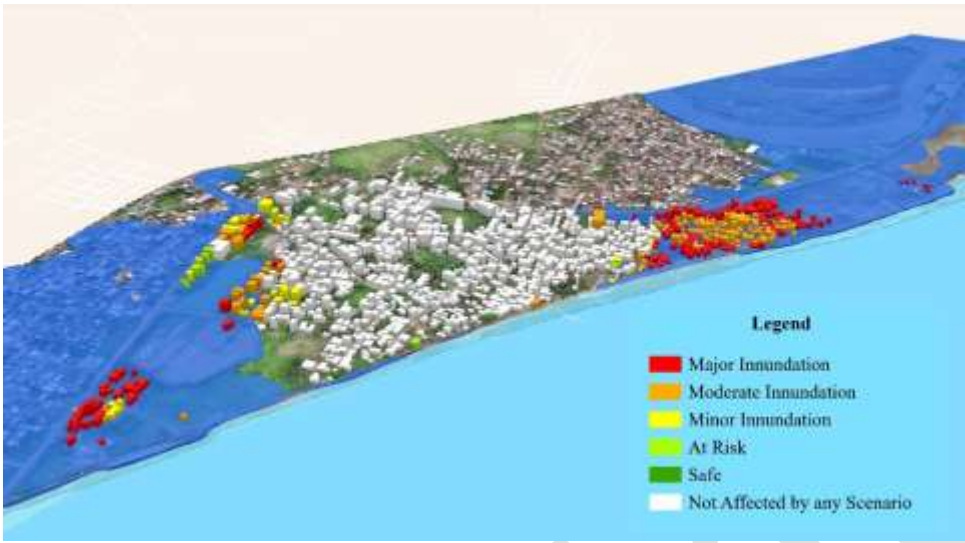


Fig. 9 3-Dimensional representation of the Flood exposure Scenario of 5.1m over the Uppada Region

4.2 Structural Typology-Based Assessment of Flood Exposure

To evaluate the vulnerability of different building structure types under varying floor exposure scenarios, four structural type categories—concrete, tile, asbestos sheds, and huts—were overlaid on the flood inundation exposure data. Analysis of flood exposure in relation to building structural types reveals distinct patterns across storm surge scenarios. In Scenario 1, no inundation was observed across all structures, reflecting lower flood intensities. However, as climatic intensity increased in Scenarios 2 and 3, the extent of inundation rose significantly, particularly among non-concrete structures like huts and asbestos sheet buildings. In contrast, reinforced concrete buildings and tile-roof buildings are assumed to demonstrate higher resilience, with the majority falling under the categories of “Moderate” to “Minor Inundation,” even in the most severe scenario. Huts are assumed to be the most vulnerable, with a large proportion experiencing “Major Inundation” in a 100-year return period, indicating high exposure to flood hazards, in Figs. 10. The comparative analysis highlights that the structural composition plays a crucial role in flood resilience, with permanent, concrete structures demonstrating higher safety margins, whereas temporary and lightweight materials exhibit a significant risk under future climatic extremes. These differences primarily reflect variations in building elevation and construction characteristics, rather than structural performance or damage resistance.



Fig. 10 Classification of Buildings as per type of building types in different scenarios

4 CONCLUSION

This study assesses flood exposure caused by storm surges using a static inundation (bathtub) approach, based on a digital terrain model (DTM) dataset derived from UAV data. The analysis of maximum water elevations (MWEs) across the study area indicates a noticeable inland expansion of inundation from the present to scenarios projected over a 100-year return period, which aligns with increasing surge elevations. Areas with relatively higher ground elevation exhibit reduced flood inundation depths, underscoring the critical role of fine-scale topography in controlling flood exposure patterns.

Additionally, residential structures' vulnerability to flood exposure is evaluated solely on estimated inundation depth and does not account for structural damage functions, building occupancy, or socio-economic vulnerability.

Consequently, the results represent relative flood exposure and structural susceptibility rather than quantitative damage or loss estimates. Despite these limitations, the methodology enables the identification of areas and building types that are likely to experience greater exposure under increasing storm surge intensities, thereby supporting prioritisation of risk reduction and preparedness measures. This will assist the local administration and disaster management authorities as a critical decision-support tool for formulating immediate preparedness plans to evacuate and relocate residents in these areas, which are highly vulnerable during flooding events. This study

serves as a starting point for demonstrating how UAV and geospatial technologies can support data-driven decision-making for sustainable coastal management and climate resilience, which can then be implemented to assess the vulnerability of the coast.

DECLARATION

The author(s) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AUTHOR CONTRIBUTIONS

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Software, Formal analysis and Visualisation: Senthil Balaganesh, Sondi Sudheer;

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Project administration: Satya Kiran Raju Alluri, Ramanathan V, Ramana Murthy M V, Kankara R S.

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